

SE Lab 4 Preparation: API Design with REST

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1 Task 1: REST API Definition

The table below defines the 8 REST API endpoints for the Search Engine System.

Method	URI	Object Accessed	Description
GET	/api/search	SearchAPI	Processes a search query and returns a list of results based on relevance.
POST	/api/search	SearchAPI	Submits a complex search request or logs a new search query for history.
GET	/api/results/{id}	SearchAPI	Retrieves detailed meta-data for a specific search result ID.
PUT	/api/results/{id}	SearchAPI	Updates the status or relevance feedback for a specific result (Admin).
POST	/api/crawler	CrawlerController	Triggers the crawler to start a new indexing job based on current config.
GET	/api/crawler/status	CrawlerController	Retrieves the current status (Idle/Running) and progress of the crawler.
POST	/api/sitemap	SitemapManager	Allows a Website Owner to upload a sitemap XML file for validation.
GET	/api/sitemap/{url}	SitemapManager	Checks the validation status of a previously submitted sitemap URL.

Table 1: REST API Definition Table for Search Engine System

2 Task 2: Class Diagram with REST API

The Class Diagram below has been updated to show which objects act as REST resources.

- We added the <<REST_API>> stereotype to **SearchAPI**, **CrawlerController**, and **SitemapManager**.

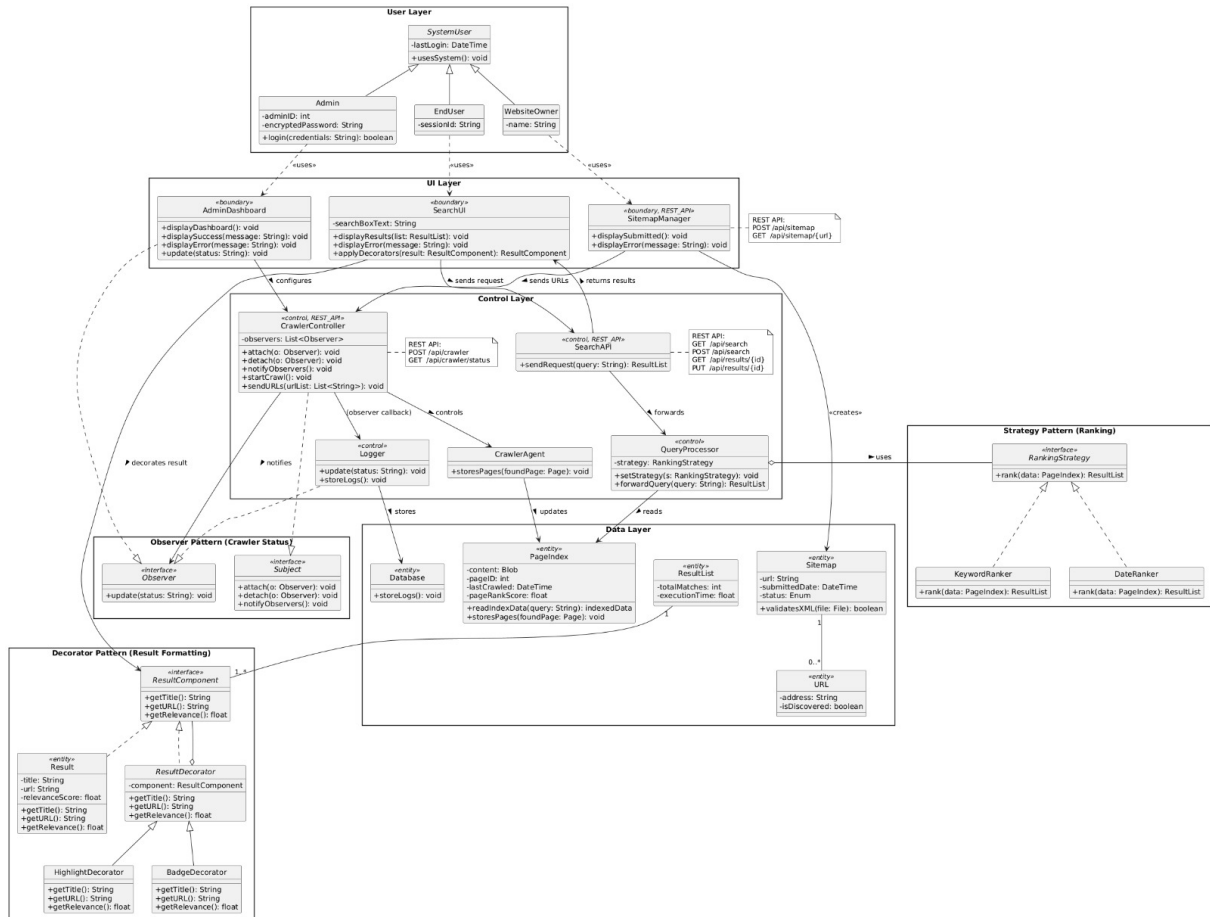


Figure 1: Updated Class Diagram identifying REST resources.

3 Task 3: Sequence Diagrams

3.1 End User Search

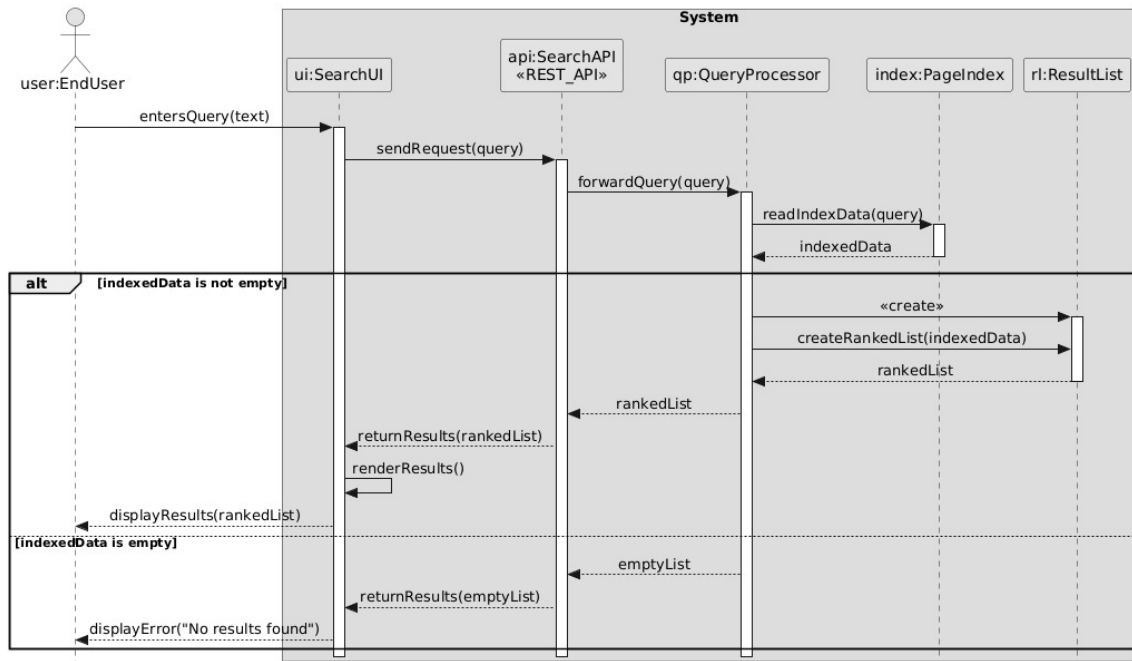


Figure 2: Sequence Diagram: Search Query via REST.

3.2 Website Owner Sitemap

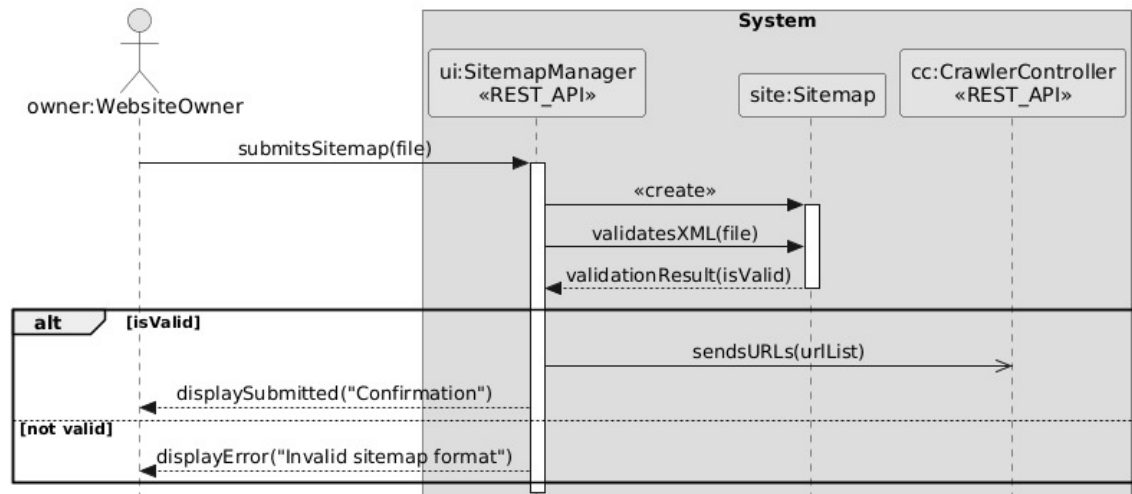


Figure 3: Sequence Diagram: Sitemap Submission via REST.

3.3 Admin Configuration

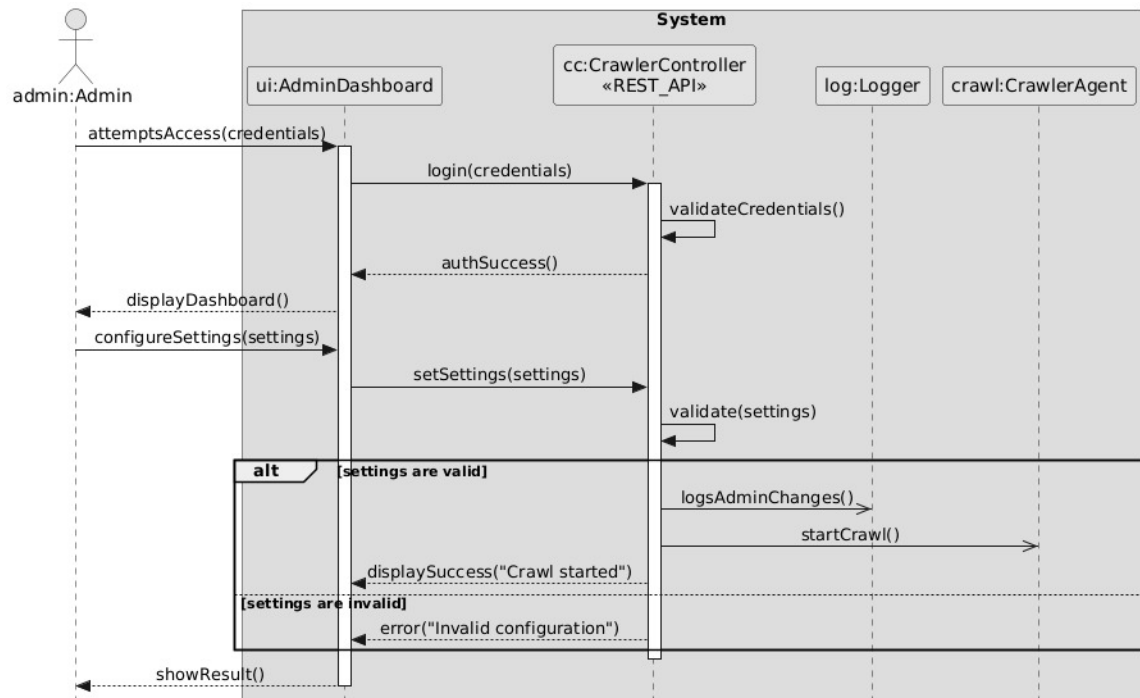


Figure 4: Sequence Diagram: Crawler Configuration via REST.